

Computational Validation of Adaptive Interface Control in TMI Quantum Systems

Extended v4 report on Tests 1-27

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Abstract

This report extends the TMI computational validation line to Tests 1-27. Version v4 adds the external simulator branch: raw Qiskit-Aer failure, Qiskit noise calibration, calibrated Qiskit-Aer validation, and circuit-inverse correctness audit. The narrow engineering chain remains:

$$I_Q^{coh-auto} \Rightarrow T_2^{eff} \uparrow \Rightarrow P_{success} \uparrow.$$

The new external-simulator result is:

$$Calibrate(I_{sim}) \wedge Correct(I_{circuit}) \Rightarrow P_{TMI} > P_{standard} > P_{base}.$$

This is not a laboratory proof of a physical field. It is reproducible computational and Qiskit-Aer simulator validation of an applied adaptive-interface control mechanism.

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1 Report status

TMI Quantum Computational Validation v0.4

The main scientific value of v4 is that the external Qiskit-Aer transfer first failed, was then calibrated, and finally was cleaned from a circuit-inverse error.

2 Summary of Tests 1-27

Test	Name	Criterion / meaning	Result
1	Ramsey toy model	$T2_TMI > T2_standard$	Delta T2 = 58.1 us
2	Noisy Ramsey learning	$T2_TMI > T2_standard$	Delta T2 = 39.3 us
3	Small quantum circuit	$P_TMI > P_standard$	Delta P = 0.115
4	Stabilized interface	Stable adaptive control	Delta P = 0.026
5	Surrogate Monte Carlo	$E[P_TMI] > E[P_standard]$	Delta = 0.0510, Pr = 0.864
6	Ablation study	Structured adaptive control	full TMI Delta = 0.0680
7	Density-matrix Monte Carlo	Full matrix validation	Delta = 0.0930, Pr = 0.933
8	Cross-model validation	Multiple circuit families	Delta = 0.0660, Pr = 0.846
9	Failure-boundary map	Weak/failure regions	failure = 0, weak = 6
10	Adversarial failure	Forced failure modes	failure cells = 58
11	Safety recovery	Mitigation after failure	recovered = 31/58
12	Reproducibility harness	One-command validation	passed = 11/11
13	Independent simulator	Independent simulator validation	Delta = 0.1121, Pr = 1.000
14	Noise-model stress	Noise class robustness	passed = 3/4, Delta = 0.0068
15	Amplitude damping recovery	Relaxation-aware layer	Delta_relax = 0.1532
16	Noise classifier	Classify then control	accuracy = 0.952, Delta = 0.0554
17	Online noise drift	Dynamic reclassification	Delta = 0.0255, acc = 0.816
18	Delayed feedback	Latency hurts online control	online = 0.883, delayed d2 = 0.861

Test	Name	Criterion / meaning	Result
19	Latency policy optimization	Predictive latency layer	opt = 0.875 > d2 = 0.861
20	Cost-overhead tradeoff	Utility and efficiency	U_highperf = 0.0705
21	Cost-interface boundary	B_cost	lambda* = 0.0636
22	Multi-objective control	Boundary manifold	standard share = 0.608, cost-aware-low = 0.373
23	Standard simulator protocol	Backend-independent Qiskit-style protocol	Delta = 0.0414, Pr = 0.750
24	Raw Qiskit-Aer execution	External simulator without calibration	failed, Delta = -0.00097, cells = 0/16
25	Qiskit noise calibration	Simulator-interface calibration	54 candidate calibrations; Loss_sim = 0.00106
26	Calibrated Qiskit-Aer validation	Held-out calibrated validation	passed, Delta = 0.0725, cells = 12/16
27	Circuit-inverse audit	Correct inverse and re-run Qiskit-Aer	passed, Delta = 0.1144, cells = 16/16

3 External simulator branch: Tests 23-27

Test 23 created a backend-independent standard-simulator protocol. The key update in v4 is the real Qiskit-Aer branch:

Test 24 : failed → Test 25 : calibration → Test 26 : calibrated passed → Test 27 : correct

3.1 Test 24: raw Qiskit-Aer failure

$$P_{base} = 0.7535, \quad P_{standard} = 0.7532, \quad P_{TMI} = 0.7523.$$

$$\Delta(P_{TMI} - P_{standard}) = -0.00097, \quad Pr(P_{TMI} > P_{standard}) = 0.0729.$$

This established the need for simulator-interface calibration:

$$I_{sim} : TMI \text{ model} \rightarrow Qiskit \text{ NoiseModel}, \quad Loss_{sim}(I_{sim}) > 0.$$

3.2 Test 25: Qiskit noise calibration

Test 25 found 54 candidate calibrations. In the default-like calibrated regime:

$$\Delta(P_{standard} - P_{base}) = 0.0191, \quad \Delta(P_{TMI} - P_{standard}) = 0.0299.$$

It also found incomplete idle-gate preservation:

$$idle_transpile_id_preservation_rate = 0.4375.$$

3.3 Test 26: calibrated Qiskit-Aer validation

$$P_{base} = 0.4793, \quad P_{standard} = 0.5203, \quad P_{TMI} = 0.5928.$$

$$\Delta(P_{standard} - P_{base}) = 0.0410, \quad \Delta(P_{TMI} - P_{standard}) = 0.0725.$$

Channels passed 4/4, but circuit-channel cells passed 12/16, pointing to a local circuit issue.

3.4 Test 27: circuit-inverse correctness audit

Test 27 found that the original circuits did not all pass ideal inverse correctness, but the corrected circuits did. The correction was:

$$S \rightarrow Sdg, \quad S^{-1} = S^\dagger.$$

After correction:

$$P_{base} = 0.5791, \quad P_{standard} = 0.6432, \quad P_{TMI} = 0.7576.$$

$$\Delta(P_{standard} - P_{base}) = 0.0641, \quad \Delta(P_{TMI} - P_{standard}) = 0.1144.$$

$$Pr(P_{standard} > P_{base}) = 1.0, \quad Pr(P_{TMI} > P_{standard}) = 1.0, \quad cells = 16/16.$$

4 External Qiskit-Aer visual summary

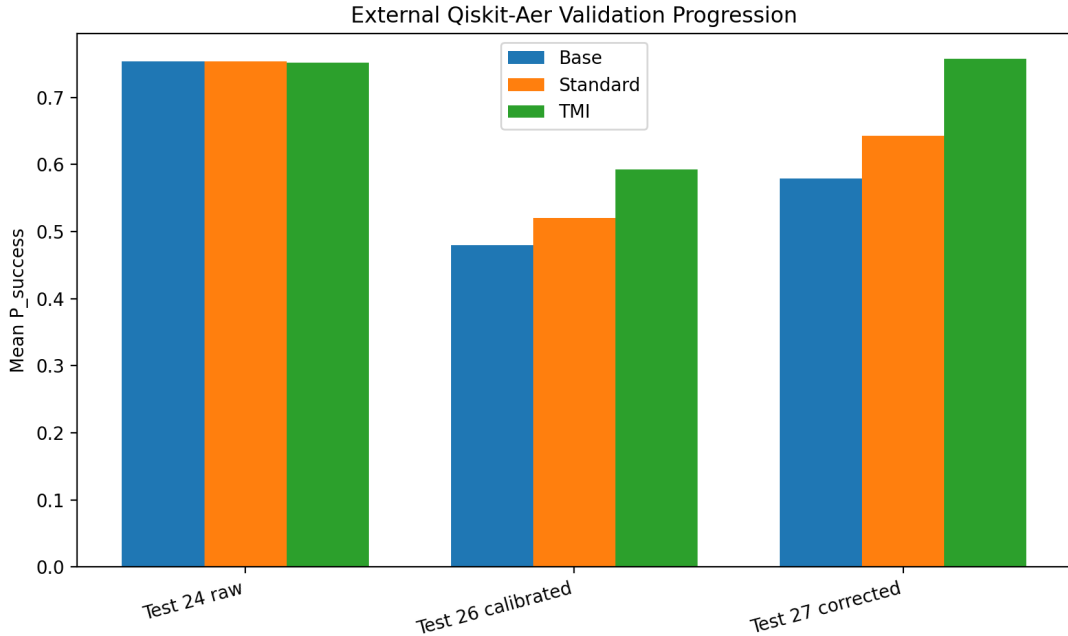


Figure 1: Mean success probabilities in Tests 24, 26 and 27.

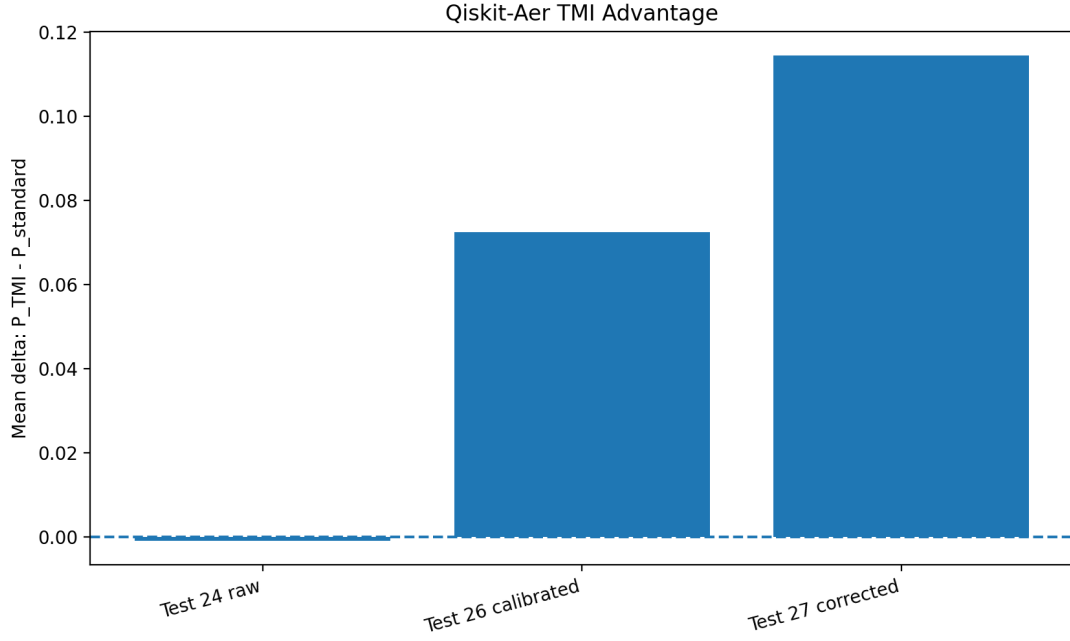


Figure 2: $P_{TMI} - P_{standard}$: negative in Test 24, positive after calibration/correction.

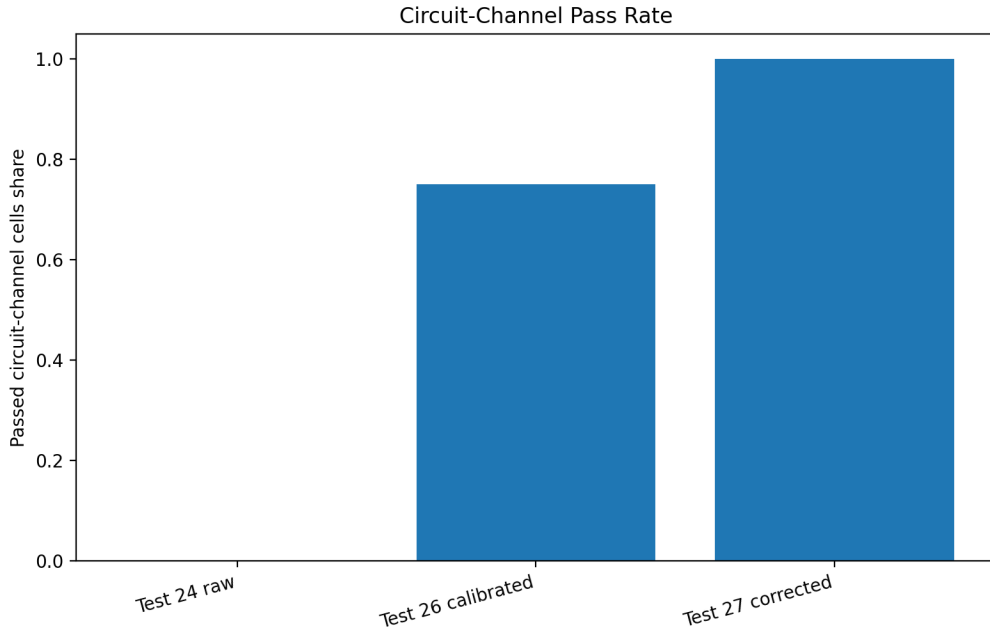


Figure 3: Passed circuit-channel cells: 0/16, 12/16, 16/16.

5 v4 formulas

$$\boxed{\text{Calibrate}(I_{sim}) \wedge \text{Correct}(I_{circuit}) \Rightarrow \text{QiskitAerValidation}}$$

$$\boxed{\text{Calibrate}(I_{sim}) \wedge \text{Correct}(I_{circuit}) \Rightarrow P_{TMI} > P_{standard} > P_{base}}$$

6 Conclusion

Tests 1-22 established internal computational validation and boundary-manifold control. Tests 23-27 add the external simulator branch. The crucial scientific point is that the report includes the raw external failure, diagnosis, calibration, circuit correction and final corrected validation.

A Package contents

The package contains Russian and English PDF/LaTeX sources, JSON/CSV summaries, external Qiskit-Aer branch figures, Tests 12-27 packages, and Tests 24-27 status files.